

4.4 Nuclear decay

This topic covers radioactive decay.

This topic may be taught using applications that relate to, for example, medical physics.

Students will be assessed on their ability to:		Suggested experiments
114	show an awareness of the existence and origin of background radiation, past and present	Measure background count rate
115	investigate and recognise nuclear radiations (alpha, beta and gamma) from their penetrating power and ionising ability	Investigate the absorption of radiation by paper, aluminium and lead (radiation penetration simulation software is a viable alternative)
116	describe the spontaneous and random nature of nuclear decay	
117	determine the half lives of radioactive isotopes graphically and recognise and use the expressions for radioactive decay: $dN/dt = -\lambda N$, $\lambda = \ln 2/t_{1/2}$ and $N = N_0 e^{-\lambda t}$	Measure the activity of a radioactive source Simulation of radioactive decay using, for example, dice
118	discuss the applications of radioactive materials, including ethical and environmental issues	